

Po-Sheng Cheng

401-432-5576

bencer@outlook.com

Portfolio: <https://bencer3283.github.io/art/>

Master of Industrial Design, Rhode Island School of Design, Jun. 2026

Cross-enrollment at School of Engineering, Brown University

B.Sc. Bio-Mechatronics Engineering, National Taiwan University, Jan 2023

Experiences

Faculty of Record, Dept. of Industrial Design, RISD, Jan. - Feb. 2026 Providence, RI

- Proposed, awarded and taught an introductory studio class (ID1531) in physical computing and embedded system development. Employed theoretical lectures, demo, and hands-on workshop to cover topic like microcontroller, circuit theory, binary arithmetic, serial communication, and machine learning.
- Blended technical skill with discussion on the societal impact of technology with framework like Science and Technology Studies.

Design Technologist Intern, Smart Design, Jun. - Aug. 2025 Brooklyn, NY

- Developed embedded firmware (STM32) for prototyping the HMI of a domestic product, while working with UX design to test and fine-tune user interaction flow and hardware user interface.
- Developed testing units for motor and pneumatic mechanisms in the product, including the electrical hardware, embedded firmware and host software. (PyQt)

Product Development Apprentice, Google, Jun. - Sep. 2024 Taipei, Taiwan

- Led a team of 5 to design a Hardware-as-a-Service platform for managing lost items in public facilities. Laid out the system architecture of the hardware kiosk, user-facing app and web services.
- My prototypes of Raspberry Pi and ESP32 hardware, a React-based website, a mobile app and a RESTful API service drove the development direction from a single hardware product to an integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022 Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating an Arduino microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.
- The prototype received widespread praise across many departments (pm, engineering, design) in the company while I worked with them to support guiding the direction of product development.

Publications

Frictional Intelligence

Posheng Cheng, NeurIPS Creative AI Track 2026, <https://openreview.net/forum?id=IRrg0HIyyR>

- Proposed novel physical interfaces of LLMs to emphasize the importance of human agency, deliberation and literacy in human-AI interaction.
- Designed, debugged and programmed several HMI embedded systems that incorporate BLDC motor, ToF sensor, touch sensing and USB-PD.

TeleSHift: Telexisting TUI for Physical Collaboration & Interaction

Andrew Chen, Tzu-Ling Yang, Shu-Yan Cheng, **Po-Sheng Cheng**, Tzu-Han Lin, and Kaiyuan Lin. 2022. This work recieved the **Best Demo Award** at **Ubicomp/ISWC 2022 Conference**, <https://doi.org/10.1145/3544793.3560323>

- Architected an ESP32-based embedded system to implement a shape-transforming device with a 3D tangible user interface.
- Designed the robust interactive mechanisms (SolidWorks). My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.

Awards

RISD Grad Common Grant 2025

awarded by the Rhode Island School of Design for the continuation of the MITM Router project.

Best Demo Paper at 2022 ACM Ubicomp conference

awarded for our paper TeleSHift: Telexisting TUI for Physical Collaboration & Interaction.

College Student Research Creativity Award

awarded by the National Science and Technology Council of Taiwan for a research project building novel spectral mapping technologies.